

$$\eta_{TS} = 0.7 = \frac{1 - \frac{T_{02}}{T_{01}}}{1 - \left(\frac{p_2}{p_{01}}\right)^{\frac{\gamma-1}{\gamma}}} = \frac{1 - \frac{T_{02}}{T_{01}}}{1 - \left(\frac{0.14}{2}\right)^{\frac{1.3-1}{1.3}}}$$

$$\Rightarrow \frac{T_{02}}{T_{01}} = 0.6790$$

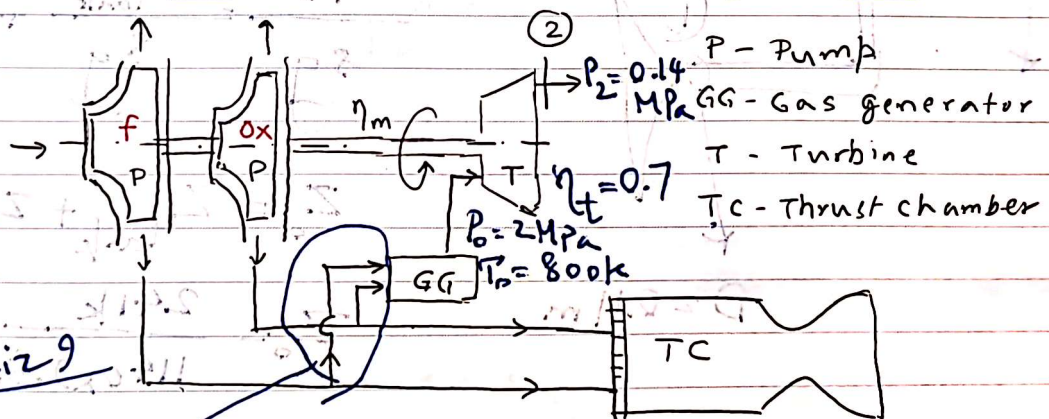
$$T_{02} = 0.6790 T_{01} = 0.6790 (800) = 543.2 \text{ K}$$

$$P_T = \dot{m}_T (1060) \left(\frac{T_{01}}{800} - \frac{T_{02}}{543.2} \right) = 272208 \dot{m}_T$$

$$\eta_m P_T = P_p \Rightarrow P_p = 0.6 \text{ MW} = \eta_m P_T = 0.9 \times$$

$$(0.90) (272208 \dot{m}_T) = 6(10^5) \Rightarrow \dot{m}_T = 2.45 \frac{\text{kg}}{\text{s}}$$

$$\text{Fraction} = \frac{\dot{m}_T}{\dot{m}} = \frac{2.45}{92} = 0.0266, \text{ or } 2.66\%$$



Same as Quiz 9

$$P_p = 0.6 \text{ MW}$$

pumps feed the Thrust chamber

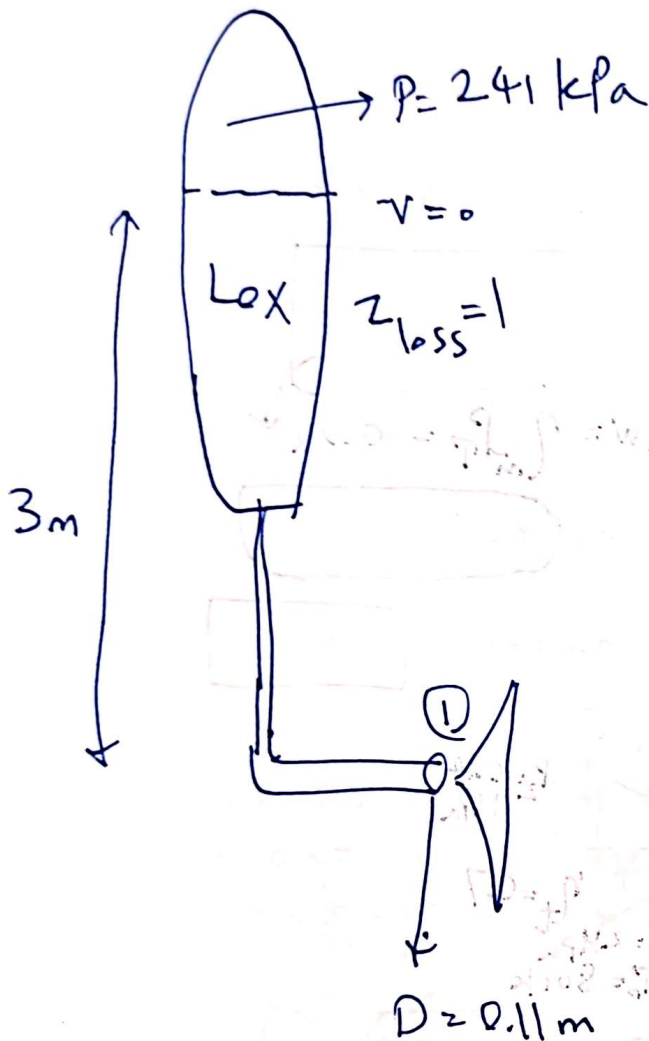
Question: $\frac{\dot{m}_{GG}}{\dot{m}_{\text{total}}}$ → How much it is.

$$\bar{M} = 34, \gamma = 1.3$$

start from η_T → get T_{02} → get turbine power

as a function of $\dot{m}_p$ put $\eta_m P_T = P_p$ → get $\dot{m}_p$

$$h = \frac{P}{\rho g}$$



I want to compute P_o of this tank

$$P_o = P_{\text{tank}} + \rho g Z - \rho g Z_{\text{loss}}$$

$\downarrow$ total $\downarrow$ static $\downarrow$ weight pressure $\downarrow$ viscous effects

$$\frac{P_o}{\rho g} = \frac{P_{\text{tank}}}{\rho g} + Z - Z_{\text{loss}}$$

$$Z_o = Z_{\text{tank}} + Z - Z_{\text{loss}}$$

$$Z_o = \frac{241 \text{ k}}{1140 \times 9.81} + 3 - 1 = \underline{\underline{23.55 \text{ m}}}$$

How to compute $V_1 = ?$

$$\text{I have } \dot{m} = 182 \frac{\text{kg}}{\text{s}} = \rho A V_1 = 1140 \left(\pi \frac{0.11^2}{4} \right) V_1$$

$$\Rightarrow V_1 = 16.8 \frac{\text{m}}{\text{s}}, \text{ How to get } P_1 = ?$$

$$\text{Bernoulli says: } \left[P_o = P_1 + \frac{1}{2} \rho V_1^2 \right] / \rho g$$

$$\Rightarrow Z_o = Z_1 + \frac{V_1^2}{2g} \Rightarrow Z_1 = Z_o - \frac{V_1^2}{2g} = \underline{\underline{9.16 \text{ m}}}$$

$$= \frac{\cancel{T_{01}} \left(1 - \frac{T_{02}}{T_{01}}\right)}{\cancel{T_{01}} \left(1 - \frac{T_{2s}}{T_{01}}\right)} = \frac{\left(1 - \frac{T_{02}}{T_{01}}\right)}{1 - \left(\frac{p_{2s}}{p_{01}}\right)^{\frac{\gamma-1}{\gamma}}} = \frac{1 - \frac{T_{02}}{T_{01}}}{1 - \left(\frac{p_2}{p_{01}}\right)^{\frac{\gamma-1}{\gamma}}}$$

Turbine Equations:

$$\underbrace{P_T = \dot{m}_T w_T}, \quad \underbrace{w_T = c_p (T_{01} - T_{02})}, \quad \boxed{\eta_{TS} = \frac{1 - \frac{T_{02}}{T_{01}}}{1 - \left(\frac{p_2}{p_{01}}\right)^{\frac{\gamma-1}{\gamma}}}} \quad \left\{ \begin{array}{l} \text{See} \\ \text{equation} \\ \text{above} \\ \text{Eqn (13.13)} \end{array} \right.$$

The turbine wheel is turned by the applied pressure ratio (p_{01}/p_{02}) across it. The turbine power is used to turn the

fuel and oxidizer pumps. ~~$P_T = \dot{S}_p$~~ Or $\boxed{P_p = \eta_m P_T}$

Ex: A turbine, powered by a gas generator, is used to drive the pumps of the previous example. The turbine entry stagnation conditions are 2 MPa and 800 K. The turbine discharge static pressure is 0.14 MPa. Calculate the fraction of the total propellant flow to be diverted to the gas generator. $\bar{M} = 34$, $\gamma = 1.3$, $\eta_{TS} = 70\%$. The turbine-pump shaft has mechanical efficiency (η_m) of 90%.

$$R = \frac{\bar{R}}{\bar{M}} = \frac{8314.3}{34} = 244.5 \frac{\text{J}}{\text{kg-K}}$$

$$c_p = \frac{\gamma R}{\gamma - 1} = 1060 \frac{\text{J}}{\text{kg-K}}$$

$$\dot{S}_T = \dot{m}_T w_T = \dot{m}_T \eta_{TS} w_{TS}$$

$$\frac{T_{02}}{T_{01}}$$

$$1 - \frac{T_{02}}{T_{01}}$$

P_v : vapour pressure : 101.3 kPa @ 90.3 k

Liquid OX becomes vapour if p drops to

$$101.3 \text{ kPa} \rightarrow Z_v = \frac{101.3 \text{ k}}{\rho g = 1140 \times 9.81} = \underline{9.06 \text{ m}}$$

Now compare Z_1 with $Z_v \rightarrow Z_1 > Z_v$

$$9.16 > 9.06 \rightarrow \boxed{\text{No-cavitation}}$$

"You can either compare p with P_v or Z with Z_v at pump entry"